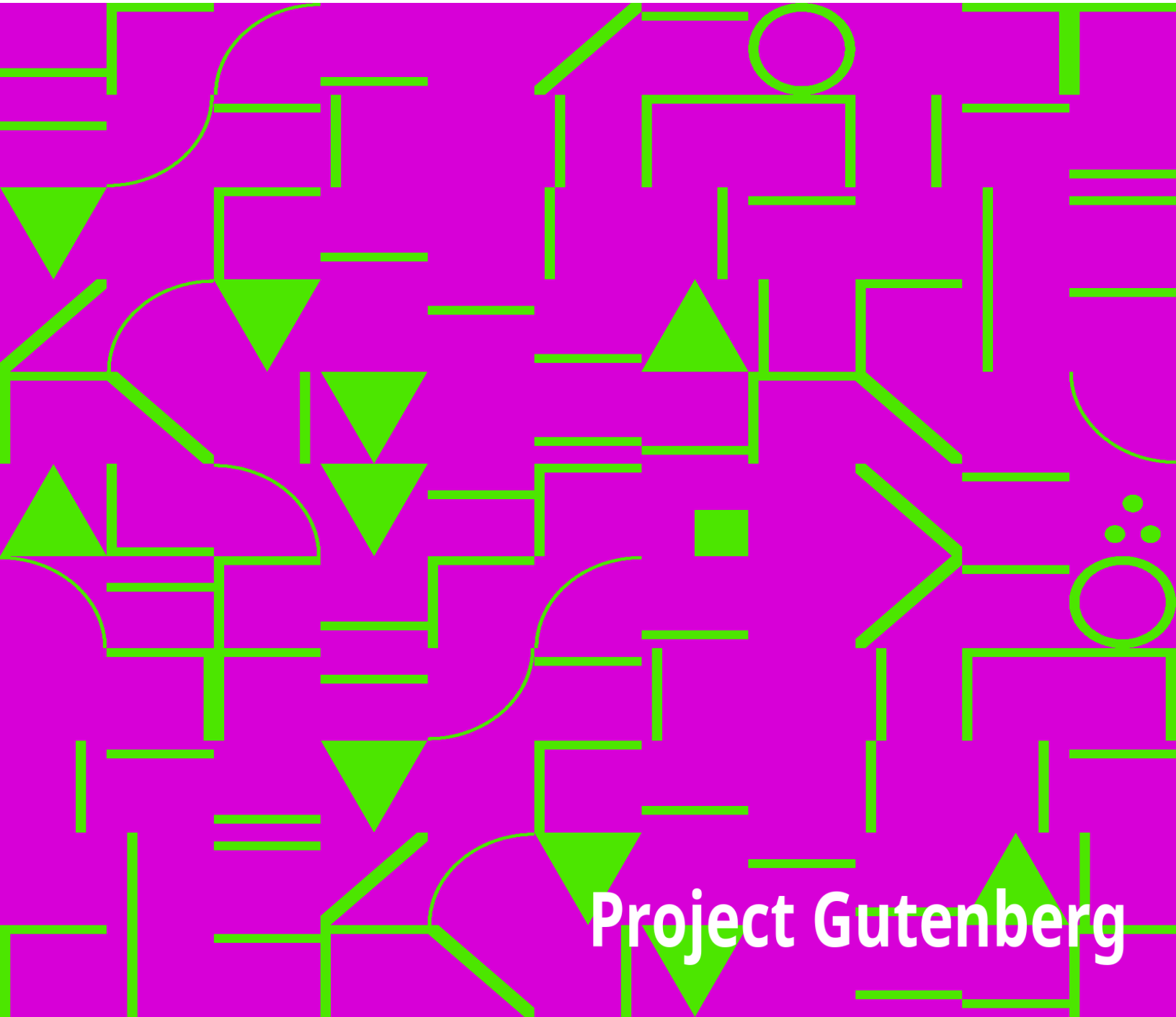


Four New Pocket Gophers of the Genus *Cratogeomys* from Jalisco, Mexico

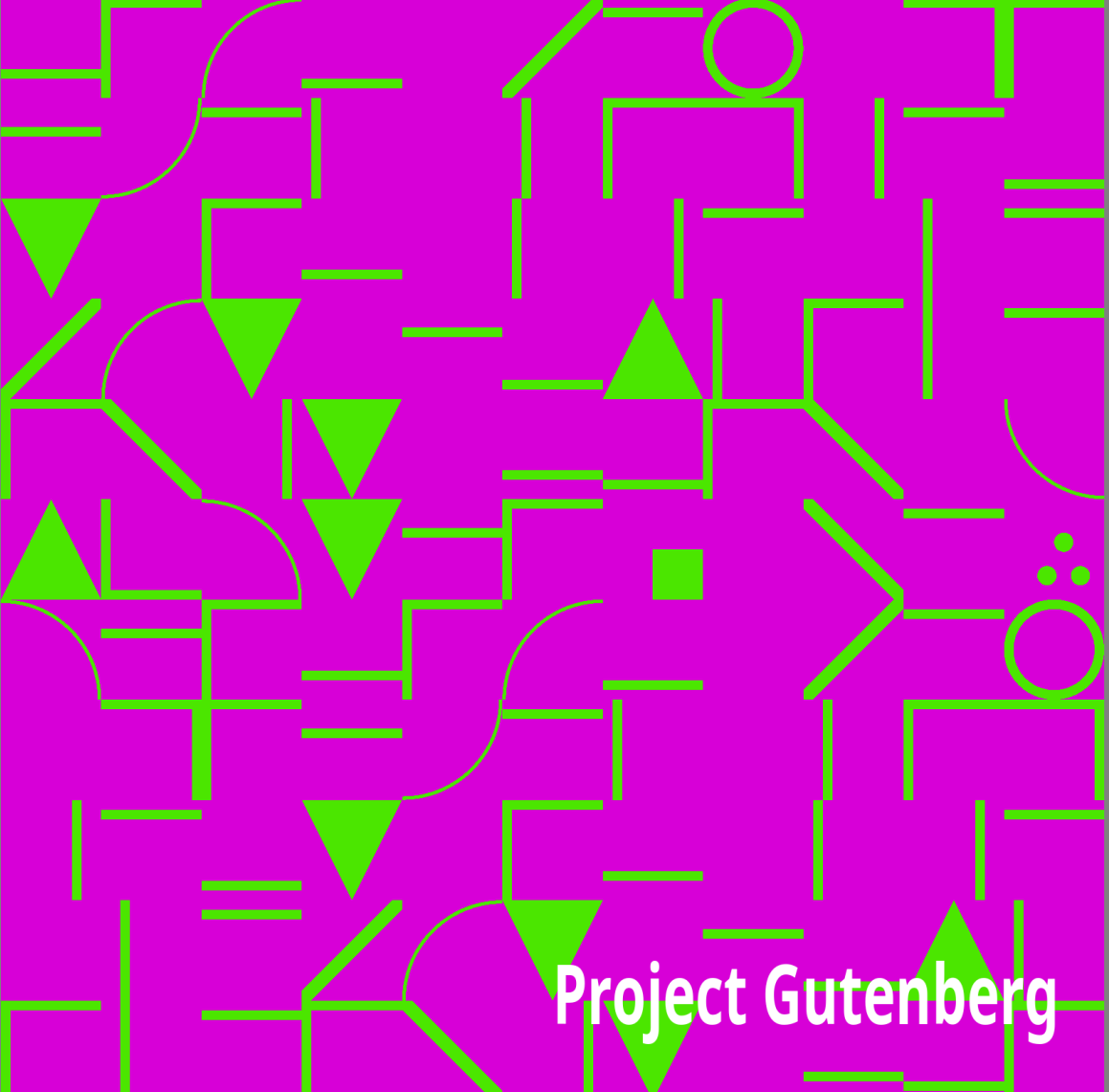
Robert J. Russell



Project Gutenberg

Four New Pocket Gophers of the Genus *Cratogeomys* from Jalisco, Mexico

Robert J. Russell

An abstract geometric pattern consisting of various shapes like squares, triangles, circles, and lines, all in shades of blue and green, creating a complex, interlocking design.

Project Gutenberg

The Project Gutenberg eBook of Four New Pocket Gophers of the Genus Cratogeomys from Jalisco, Mexico

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Four New Pocket Gophers of the Genus Cratogeomys from Jalisco, Mexico

Author: Robert J. Russell

Release date: January 31, 2010 [eBook #31141]

Most recently updated: January 6, 2021

Language: English

Other information and formats: www.gutenberg.org/ebooks/31141

Credits: Produced by Chris Curnow, Joseph Cooper, Diane Monico, and the Online Distributed Proofreading Team at <https://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK FOUR NEW
POCKET GOPHERS OF THE GENUS CRATOGEOMYS FROM
JALISCO, MEXICO ***

Four New Pocket Gophers of the Genus *Cratogeomys* from Jalisco, Mexico

BY

ROBERT J. RUSSELL

**University of Kansas Publications
Museum of Natural History**

**Volume 5, No. 31, pp. 535-542
October 15, 1953**

University of Kansas

LAWRENCE

1953

UNIVERSITY OF KANSAS PUBLICATIONS, MUSEUM OF NATURAL HISTORY

Editors: E. Raymond Hall, Chairman, A. Byron Leonard,
and Robert W. Wilson

Volume 5, No. 31, pp. 535-542
October 15, 1953

UNIVERSITY OF KANSAS
Lawrence, Kansas

PRINTED BY
FERD VOILAND, JR., STATE PRINTER
TOPEKA, KANSAS
1953

24-8662

Four New Pocket Gophers of the Genus *Cratogeomys* from Jalisco, Mexico

By

ROBERT J. RUSSELL

In the course of my taxonomic study of the genus *Cratogeomys*, a high degree of variation was found between several populations of these gophers in central Jalisco. Two species, *C. gymnurus* and *C. zinseri*, occur in this part of the state. Previously *C. gymnurus* was known only from southern Jalisco and *C. zinseri* only from extreme eastern Jalisco, but through the efforts of J. R. Alcorn specimens were obtained of both species in the central part of the state. These large gophers are difficult to collect, and I am grateful to him for securing this significant material. Costs of the field work were defrayed by the National Science Foundation and the Kansas University Endowment Association. Thanks are due also to those in charge of the United States Biological Surveys Collection for the loan of comparative material. Study of the recently acquired specimens taken in central Jalisco reveals two undescribed subspecies each of *C. gymnurus* and *C. zinseri*. These may be known and described as

***Cratogeomys gymnurus tellus* new subspecies**

Type.—Female, adult, skull and skin, No. 33454 Mus. Nat. Hist., Univ. Kansas; from 3 mi. W Tala, 4300 ft., Jalisco, México; obtained on June 2, 1949, by J. R. Alcorn, original No. 9376.

Range.—North-central Jalisco; known from several localities in the vicinity of Tala.

Diagnosis.—Size large (see measurements); tail long, naked; hind foot small; color pale for species, upper parts Kaiser Brown (capitalized terms are of Ridgway, Color Standards and Color Nomenclature, Washington, D. C., 1912), bases of individual hairs Plumbeous, tips Hazel, underparts

creamy-white, bases of hairs Plumbeous; skull large, relatively narrow, rugose; zygomatic breadth narrower posteriorly than anteriorly; rostrum shallow, relatively broad in males, narrower in females; interorbital region broad; braincase narrow and flattened; basioccipital relatively wide, especially anteriorly; mastoid processes of squamosal large, knoblike; paroccipital processes long, extending laterally over more than half the width of mastoid bullae; upper incisors projecting anteriorly; maxillary teeth relatively large.

Comparisons.—From topotypes of *C. g. gymnurus* from Zapotlan, Jalisco, the most closely related subspecies, *C. g. tellus* differs in: Body smaller (total length averaging 338 instead of 341 in females and 356 instead of 369 in males); hind foot smaller (averaging 45 instead of 50 in females and 47 instead of 51 in males); color more brownish above, creamy-white rather than buffy below; skull smaller, especially in females (basilar length averaging 55.3 instead of 57.5 in females and 57.7 instead of 60.5 in males), narrower, and more rugose; zygomatic breadth less in females (averaging 42.5 compared with 46.2), greater in males (48.0 compared with 46.7); zygomata more nearly parallel; auditory bullae relatively smaller; mastoid processes of squamosal larger, knoblike; paroccipital processes longer, extending farther laterally; rostrum less massive; upper incisors projecting anteriorly, instead of being strongly recurved; maxillary teeth relatively larger.

From near-topotypes of *C. g. inclarus* from the Sierra Nevada de Colima, Jalisco, *C. g. tellus* differs in: Hind foot smaller; color paler brownish above in contrast to glossy black, creamy-white below rather than buffy, feet clothed with whitish instead of brownish hairs; skull larger (basilar length averaging 55.3 in females compared with 53.2, no males available for comparisons); zygomatic breadth less; nasals longer, extending posterior to front edge of anterior roots of zygomata rather than ending even with, or slightly behind, them; rostrum more massive; mastoid processes of squamosal larger; paroccipital processes extending farther laterally; upper incisors projecting anteriorly, rather than recurved; maxillary teeth larger (length of maxillary tooth-row averaging 14.6 compared with 13.3).

Close comparison with *C. g. imparilis* from Pátzcuaro, Michoacán, is not needed, *but C. g. tellus* differs especially in: Color of underparts and hairs

of feet whitish rather than brownish; skull smaller; zygomatic breadth greater; interorbital constriction broader; nasals longer; maxillary tooth-row longer.

Measurements.—Averages of three adult females, including type, and the measurements of an adult male (in parentheses) from the type locality are: Total length, 338 (356); length of tail, 92 (87); length of hind foot, 45 (47); occipitonasal length of skull, 64.1 (68.7); basilar length, 55.3 (61.4); zygomatic breadth, 42.7 (48.0); interorbital breadth, 9.6 (11.4); greatest height of cranium, taken from palate perpendicular to line touching two highest points on top of skull, 23.9 (25.3); least depth of rostrum, 10.6 (11.7); breadth of rostrum, 14.7 (16.5); length of nasals, 23.6 (25.2); width across mastoid processes of squamosal, 43.8 (49.7); height of occiput, 18.1 (19.9); length of maxillary tooth-row, 14.6 (15.2).

Remarks.—The distribution of *C. gymnurus* is spotty; its occurrence seemingly depends on edaphic conditions. The isolation of soils with textures suitable to this animal has resulted in the isolation of gopher populations. The distribution is similar to that of species occurring on islands. In this instance, however, the populations of gophers are separated by soils of heavy texture which render burrowing difficult or impossible for gophers. Such conditions have led to a high degree of subspeciation in a relatively short distance. For example, four subspecies of *C. gymnurus* occur in Jalisco, and, all are within an area scarcely fifty miles in diameter.

Cratogeomys gymnurus tellus is the northernmost subspecies of *C. gymnurus*. It is known from only the vicinity of Tala, west of Guadalajara, and its range probably is not much more extensive than this because of the localized distribution of suitable soils.

Specimens examined.—Total number ten, as follows: 3 mi. W Tala, type locality, 5; 1 mi. NE Tala, 3; 1 mi. S El Refugio, 2.

***Cratogeomys gymnurus atratus* new subspecies**

Type.—Female, adult, skull and skin, No. 31880 Mus. Nat. Hist., Univ. Kansas; from Top of Cerro Viejo de Cuyutlán, 9700 ft., 19 mi. S and 9 mi.

W Guadalajara, Jalisco, México; obtained on February 17, 1949, by J. R. Alcorn, original No. 7902.

Range.—Known only from the type locality on the Cerro Viejo.

Diagnosis.—Size small (see measurements); tail long; hind foot small; color dark, upper parts glossy Blackish Brown, bases of hairs Plumbeous, sides Chestnut Brown, underparts Pale Ochraceous-Buff or Warm Buff mixed with Plumbeous of the hair bases; skull small, lightly constructed, relatively deep; zygomata relatively weak; zygomatic breadth wider posteriorly than anteriorly; rostrum relatively wide, especially in males; nasals relatively long, truncate posteriorly; interorbital constriction narrow; braincase inflated; mastoid processes of squamosal only slightly wider than zygomatic breadth; auditory bullae relatively large; paroccipital processes weakly constructed, but extend laterally over half the width of mastoid bullae; upper incisors projecting anteriorly, rather than being strongly recurved; maxillary teeth small.

Comparisons.—From topotypes of *C. g. gymnurus* from Zapotlan, Jalisco, *C. g. atratus* differs in: Body smaller (total length averaging 300 in females compared with 341, a male measured 315 compared with an average of 363); tail shorter, hind foot smaller; color of upper parts darker, glossy Blackish-Brown rather than reddish brown, underparts paler; skull smaller (basilar length averaging 48.6 compared with 57.5, a male measured 50.0 compared with an average of 59.0); zygomata more weakly constructed; zygomatic breadth less, and wider posteriorly than anteriorly; braincase more inflated; nasals shorter; rostrum relatively narrower and shallower; width across mastoid processes of squamosals less; paroccipital processes less strongly constructed, extending farther laterally; upper incisors projecting anteriorly rather than being strongly recurved; maxillary teeth smaller (length of maxillary tooth-row averaging 11.2 compared with 14.9).

From *C. g. tellus*, that occurs to the northwest, *C. g. atratus* differs in: Body smaller; hind foot slightly smaller; upper parts darker; underparts Pale Ochraceous-Buff rather than creamy-white; skull smaller (see measurements); zygomatic breadth less, and wider posteriorly than anteriorly; nasals shorter, truncate posteriorly rather than emarginate; rostrum narrower and shallower; maxillary teeth smaller.

From near-topotypes of *C. g. inclarus* from the Sierra Nevada de Colima, *C. g. atratus* differs in: Body slightly smaller; hind foot smaller (averaging 42 compared with 49); color of upper parts near the same, underparts paler; skull smaller, narrower, weaker in construction; zygomatic breadth less; nasals relatively longer, but actually shorter (averaging 19.7 compared with 20.3); upper incisors projecting anteriorly rather than being recurved; maxillary teeth smaller.

Measurements.—The type and an adult female (its measurements in parentheses) yield measurements as follows: Total length, 300 (299); length of tail, 78 (83); length of hind foot, 43 (40); occipitonasal length of skull, 56.3 (55.5); basilar length, 49.3 (47.8); zygomatic breadth, 37.9 (36.5); interorbital breadth, 8.7 (8.1); greatest height of cranium, as explained above, 21.6 (20.7); least depth of rostrum, 9.2 (8.8); breadth of rostrum, 12.8 (12.7); length of nasals, 19.4 (20.0); width across mastoid processes of squamosal, 38.2 (37.1); height of occiput, 16.9 (17.3); length of maxillary tooth-row, 11.9 (11.3).

Remarks.—*Cratogeomys gymnurus atratus* is the smallest subspecies known for the species, and is so distinct from other described subspecies, that it is difficult to select one as the closest relative. In color, *C. g. atratus* closely resembles *C. g. inclarus*, which occurs at comparable elevations in the Sierra Nevada, but the skulls are unlike. Among named subspecies of *C. gymnurus*, the skull of *tellus* most closely resembles that of *atratus*, and, although they differ greatly in size, *C. g. tellus* seems to be the closest relative of *C. g. atratus*.

This newly described subspecies is known only from Cerro Viejo and is probably restricted to the higher elevations on this mountain.

Specimens examined.—Seven, Top of Cerro Viejo de Cuyutlán, 19 mi. S and 9 mi. W Guadalajara.

***Cratogeomys zinseri zodiuss* new subspecies**

Type.—Male, adult, skull and skin, No. 31879 Mus. Nat. Hist., Univ. Kansas; from 13 mi. S and 15 mi. W Guadalajara, Jalisco, México; obtained on February 6, 1949, by J. R. Alcorn, original No. 7747.

Range.—Known only from the type locality.

Diagnosis.—Size small (see measurements); tail short; hind foot small; upper parts Sayal Brown, underparts Pinkish Buff, hind foot whitish; skull small, narrow; outline of dorsal profile of skull concave; zygomatic breadth narrow; nasals actually short, but relatively long; width across mastoid processes of squamosal short; auditory bullae inflated; interorbital constriction narrow; maxillary teeth relatively large.

Comparisons.—From topotypes of *C. z. zinseri* from Lagos, Jalisco, *C. z. zodioides* differs in: Body smaller (see measurements); tail shorter, hind foot smaller; upper parts dull brownish instead of reddish-brown, underparts paler, hairs of feet whitish instead of brownish; skull smaller, especially in females, narrower; dorsal profile of skull concave or flat (females) rather than convex; zygomatic breadth less; rostrum narrower and shallower; nasals actually shorter, but relatively longer in relation to length of skull; width across mastoid processes of squamosal shorter; maxillary teeth smaller.

Measurements.—The type and an adult male (his measurements in parentheses) yield measurements as follows: Total length, 318 (324); length of tail, 95 (89); length of hind foot, 41 (41); occipitonasal length of skull, 60.5 (59.1); basilar length, 52.4 (51.8); zygomatic breadth, 40.6 (39.0); interorbital breadth, 8.3 (8.8); greatest height of cranium, as explained above, 22.6 (22.4); least depth of rostrum, 11.2 (10.4); breadth of rostrum, 13.3 (13.9); length of nasals, 21.7 (21.2); width across mastoid processes of squamosal, 37.1 (36.8); height of occiput, 17.7 (17.9); length of maxillary tooth-row, 13.0 (13.3). A nearly adult female measured: Total length, 292; length of tail, 81; length of hind foot, 39; occipitonasal length of skull, 53.3; basilar length, 46.5; zygomatic breadth, 34.1; interorbital breadth, 7.8; greatest height of cranium, 21.0; least depth of rostrum, 9.8; length of nasals, 18.0; width across mastoid processes of squamosal, 32.2; depth of occiput, 15.9; length of maxillary tooth-row, 12.1.

Remarks.—This newly described subspecies is the smallest of known races of *C. zinseri*, and it is seemingly more closely related to *C. z. zinseri* than to the subspecies newly named below from the north end of Lago Sayula. The skulls of females are especially small and delicate in structure; the males

are larger with more massive skulls. *C. z. zodijs* is known to occur in the foot hills north of the Cerro Viejo, the mountain from which *C. g. atratus* was described above.

Specimens examined.—Seven, 13 mi. S and 15 mi. W Guadalajara.

***Cratogeomys zinseri morulus* new subspecies**

Type.—Male, adult, skull and skin, No. 36679 Mus. Nat. Hist., Univ. Kansas; from N end Lago Sayula, 4400 ft., 9 mi. N and 2 mi. E Atoyac, Jalisco, México; obtained on March 23, 1950, by J. R. Alcorn, original No. 10889.

Range.—Known only from the type locality in central Jalisco.

Diagnosis.—Size large (see measurements); tail short; hind foot large; upper parts Fuscous-Black, strongly mixed with Walnut Brown, underparts Cinnamon-Buff, bases of all hairs Plumbeous; skull large, broad, rugose; outline of dorsal profile slightly concave, almost flat; zygomata strongly constructed, maxillary arm almost touching squamosal arm over jugal; wide across zygomata; nasals actually and relatively long; rostrum relatively narrow; auditory bullae inflated, relatively large; maxillary teeth relatively large.

Comparisons.—From topotypes of *C. z. zinseri* from Lagos, Jalisco, *C. z. morulus* differs in: Tail shorter (averaging 96 in females compared with 101, 94 in males compared with 115); color darker above, Fuscous-Black instead of Cinnamon-Rufous, underparts paler; skull larger (occipitonasal length 63.7 rather than averaging 58.5 in females and 68.6 rather than 63.1 in males); wide across zygomata; nasals actually and relatively longer; rostrum relatively narrower; wider across mastoid processes of squamosal; auditory bullae inflated, relatively larger; maxillary teeth larger.

From *C. z. zodijs*, that occurs to the northeast, *C. z. morulus* differs in: Body larger (see measurements); hind foot larger; color of upper parts darker, underparts paler; skull much larger, broader, more rugose; dorsal profile of skull slightly concave rather than convex; wider across zygomata; nasals actually and relatively longer; rostrum broader, more massive; wider

across mastoid processes of squamosal; auditory bullae larger; maxillary teeth larger.

Measurements.—The type and an adult female (her measurements in parentheses) from the type locality measure: Total length, 358 (338); length of tail, 94 (97); length of hind foot, 49 (45); occipitonasal length of skull, 68.6 (63.7); basilar length, 58.0 (55.6); zygomatic breadth, 49.3 (45.0); interorbital breadth, 9.6 (8.9); greatest height of cranium, as explained above, 26.1 (24.6); least depth of rostrum, 12.5 (10.8); breadth of rostrum, 14.5 (13.7); length of nasals, 25.9 (22.5); width across mastoid processes of squamosal, 47.7 (42.8); height of occiput, 19.8 (17.8); length of maxillary tooth-row, 13.9 (13.8).

Remarks.—*Cratogeomys zinseri morulus* is the darkest subspecies known of *C. zinseri*. It differs widely from other subspecies of this species in color and the large size of the skull.

Cratogeomys zinseri occurs over the same general area as *C. gymnurus* in central Jalisco, although these two species seemingly do not share the same local habitat. *C. zinseri* differs from *C. gymnurus* as follows: Tail relatively longer; skull wider across zygomatic arches than across mastoid processes of squamosal (reverse true in *C. gymnurus*); zygomata strongly bowed outward anteriorly; maxillary arm of zygomata almost touching squamosal arm (instead of widely separated from each other) above jugal; rostrum relatively narrower, less massive; border of nasals parallel or laterally swollen instead of gradually tapering.

Specimens examined.—Four, N end of Lago Sayula, 9 mi. N and 2 mi. E Atoyac.

Museum of Natural History, University of Kansas, Lawrence. Transmitted June 12, 1953.

24-8662

Transcriber's Notes:

Page 538: Changed lead to led (conditions have lead to a high degree).

*** END OF THE PROJECT GUTENBERG EBOOK FOUR NEW
POCKET GOPHERS OF THE GENUS CRATOGEOMYS FROM
JALISCO, MEXICO ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.